

MATH 114 - Computational Stochastics

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Contents

1	Overview	1
1.1	Introduction	1
1.2	Sampling	2
1.3	MCMC	2
2	Markov Chains	2
2.1	Lecture 9	2
2.2	Lecture 10	4

1 Overview

1.1 Introduction

Stochastic models describe real life phenomena like motion of the cell, shape of polymers stock prices, and more. What can we compute analytically and numerically from these models?

Key questions:

1. What is the distribution of the system?
2. How can we sample from the distribution?
3. How can we compute mean, variance, and higher moments of these systems?

Main topics:

1. Sampling RVs
2. Markov chain Monte Carlo (MCMC)
3. Stochastic differential equations (numerical & computation)

1.2 Sampling

- Generating "hard" RVs from "easy" RVs.
- Compute "hard" integrals.

ex Area of the unit circle, $A = \pi$

Let $U_1 \sim U(-1, 1)$ and $U_2 \sim U(-1, 1)$

Let $x = \begin{cases} 0 & \text{if } (u_1, u_2) \text{ outside circle} \\ 1 & \text{if } (u_1, u_2) \text{ inside circle} \end{cases}$

$\implies P(X = 1) = \frac{A}{4}, P(X = 0) = 1 - \frac{A}{4}$

$\mathbb{E}(X) = \frac{A}{4}$

After N trials we generate $X_1, X_2, \dots, X_n$ iid

By the strong law of large numbers

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N X_i = \frac{A}{4} = \frac{\pi}{4}$$

1.3 MCMC

Construct Markov chains to generate target distribution.

ex Ising model

2 Markov Chains

2.1 Lecture 9

Def: A stochastic process is a family of RVs, $X_t \in S$, indexed by t , where $S \subseteq \mathbb{R}^d$ is a state space. Typically, $t \in [0, \infty)$, or $t \in [a, b]$, or $t \in \{0, 1, 2, 3, \dots\}$, etc.

- A discrete time stochastic process, is a sequence of RVs, $X_n \in S$, $n = 0, 1, 2, \dots$ or $n = 1, 2, 3, \dots$
- If S is finite or countably infinite then the stochastic process is a discrete state process.

First, we will examine discrete-time, discrete pace, stochastic process.

Two basic questions:

- Find the joint distribution

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) \\ i_0, i_1, \dots, i_n \in S$$

- Find the long-time behavior of the distribution of X_n :

$$P(x_n = i), i \in S, n \gg 1$$

Def: Markov processes

- A stochastic process $\{X_n\}_{n=0}^{\infty}$ is a Markov process if

$$P(X_{n+1} = i_{n+1} | x_0 = i_0, \dots, x_n = i_n) \\ = P(X_{n+1} = i_{n+1} | X_n = i_n) \\ \forall n \geq 0, \forall i_0, \dots, i_{n+1} \in S$$

- A Markov process is time homogeneous if:

$$P(x_{n+1} = j | x_n = i) \\ = P(X_1 = j | X_0 = i) \\ \forall n \geq 0, \forall i, j \in S$$

Here we consider discrete time/state, time homogeneous Markov Chains (MC).

Def: Transition probability

$$P(i, j) = P(X_1 = j | X_0 = i) \\ = P(i \rightarrow j) \quad \forall i, j \in S$$

Def: Transition matrix

$$P = [p(i, j)] \text{ if } S = \{1, 2, 3, \dots\} \text{ or } \{0, 1, 2, \dots\}$$

Here i, j are states and also are row/column labels

For any $n \geq 1, i_0, \dots, i_n \in S$, we have:

$$\begin{aligned} P(X_0 = i_0, \dots, X_n = i_n) \\ &= P(X_0 = i_0)P(X_1 = i_1|X_0 = i_0)P(X_2 = i_2|X_1 = i_1, X_0 = i_0) \\ &\dots P(X_n = i_n|X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \quad [\text{conditional probability}] \end{aligned}$$

ex Let $X_0, X_1, \dots, X_n \in S$ iid. Then $\{X_n\}_{n=0}^\infty$ is a time homogeneous MC.
Let $S = \{1, 2, \dots, N\}$. Then $p(i, j) = \mathbb{P}(X_1 = j|X_0 = i) = \mathbb{P}(X_1 = j) = p_j$

$$\Rightarrow p = \begin{bmatrix} p_1 & \dots & p_n \\ \vdots & \ddots & \vdots \\ p_1 & \dots & p_n \end{bmatrix}, \sum_{i=1}^n p_i = 1$$

2.2 Lecture 10

Recap:

- Discrete time/state, time homogeneous
- MC: $\{X_n\}_{n=0}^\infty \in S$
- $\mathbb{P}(X_{n+1} = i_{n+1}|X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_{n+1} = i_{n+1}|X_n = i_n)$
 $= p(i_n, i_{n+1})$
- Transition probability $\mathbb{P}(X_1 = j|X_0 = i)$
 $= p(i, j) = p_{ij} \quad \forall i, j \in S$
- Transition matrix $\mathbb{P}[p(i, j)]$

ex Two-state MC, e.g. state of phone $X_n \in S = \{0, 1\}$

$$\begin{cases} X_n = 0 & \text{phone free at time } n \\ X_n = 1 & \text{phone busy at time } n \end{cases}$$

$$\mathbb{P}(\text{free} \rightarrow \text{busy in time step}) = p$$

$$\mathbb{P}(\text{busy} \rightarrow \text{free in time step}) = q$$

$$\mathbb{P}(\text{free} \rightarrow \text{free in time step}) = 1 - p$$

$$\mathbb{P}(\text{busy} \rightarrow \text{busy in time step}) = 1 - q$$

Transition matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

$$\text{e.g. } p = \frac{1}{2}, q = \frac{1}{3}$$

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

$$\text{Let } \mathbb{P}(X_0 = 0) = \frac{1}{5}, \mathbb{P}(X_0 = 1) = \frac{4}{5}$$

$$\text{Then } \mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 0)$$

$$= \mathbb{P}(X_0 = 0) \cdot p(0, 1) \cdot p(1, 0)$$

$$= \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{30}$$

$$\text{And } \mathbb{P}(X_0 = 0, X_2 = 0) = \sum_{i=0}^1 \mathbb{P}(X_0 = 0, X_1 = i, X_2 = 0)$$

$$\boxed{\text{ex}} \quad P_0 = I, p_0(i, j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$P_1 = P, p_1(i, j) = p(i, j)$$

The distribution vector π_n for X_n is

$$\pi_n = (\pi_n(0), \pi_n(1), \dots)$$

$$\text{where } \pi_n(i) = \mathbb{P}(X_n = i), i \in S$$

Note that π_n is a probability vector

Property: Let $X_n \in S = \{1, 2, 3, \dots\}, n \geq 0$ be the time homogeneous Markov Chain with transition matrix P . Then

- P is a stochastic matrix

$$\bullet \quad P \cdot \mathbb{I} = \mathbb{I} \text{ where } \mathbb{I} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

- 1 is an eigenvalue of P with eigenvector $\mathbb{I}$

- $P_{m+n} = P_m \cdot P_n$ (Chapman-Kolmogorov equation)
- $P_n = P^n$
- $\pi_n = \pi_0 P_n = \pi_0 P^n$
- $\pi_{n+1} = \pi_n P, \forall n \geq 0$

Proof: direct verification

e.g. $\mathbb{P}(X_1 = j) = \sum_{i \in S} \mathbb{P}(X_0 = i) \mathbb{P}(X_1 = j | X_0 = i)$

$\pi_1(j) = \sum \pi_0(j) P_{ij}$

Long-run behavior, invariant distribution

Def: Let $P = [p(i, j)]$ be the transition matrix for Markov Chain $\{X_n\}_{n=0}^{\infty}$ with $S = \{1, 2, \dots, N\}$. A probability vector $\pi \in \mathbb{R}^n$ is a stationary or invariant distribution if $\pi^P = \pi$